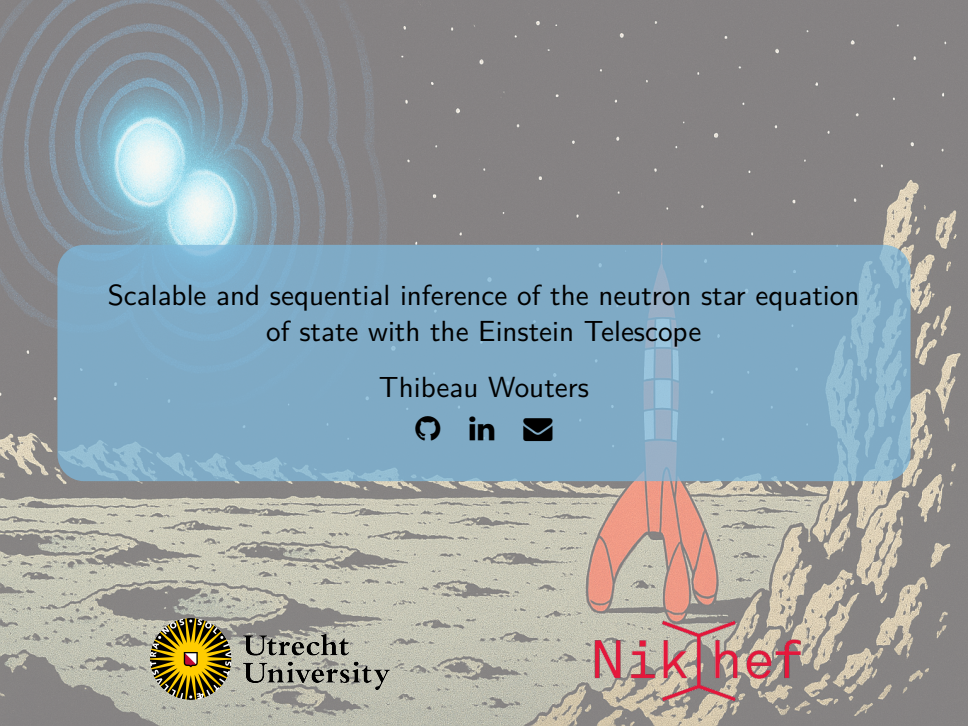




Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope

ThibEAU Wouters



Utrecht
University



Equation of state inference in the 3G era

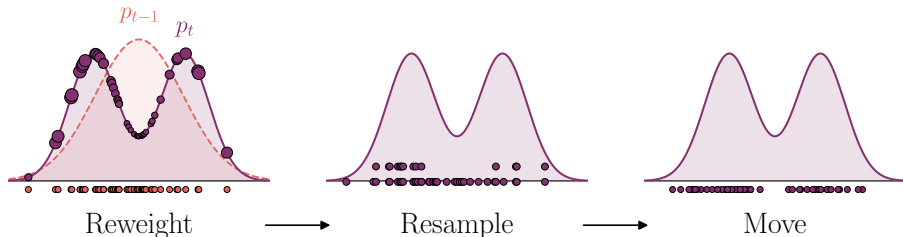
- Tidal deformations in BNS mergers probe the neutron-star equation of state (EOS)
- Einstein Telescope/Cosmic Explorer: $\mathcal{O}(10^4)$ BNS mergers per year [1]
- **Bottleneck**: existing methods must restart from scratch as the catalog grows, we need
 - Scalable: e.g. GPU-acceleration (JESTER)
 - **Sequential**: reuse previous results, avoid restarting from the prior

Goal

Natural framework for scalable, sequential inference: **sequential Monte Carlo (SMC)**

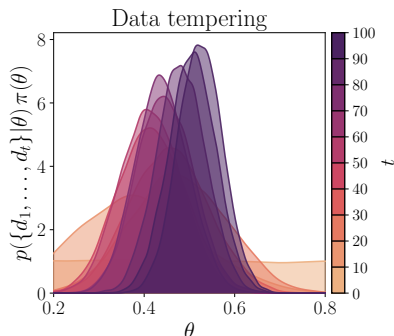
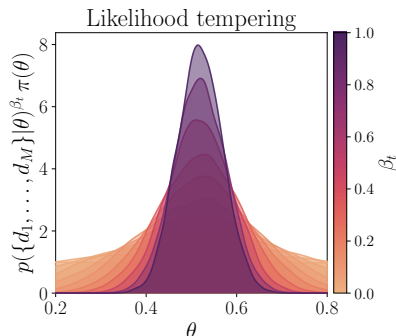
Sequential Monte Carlo (SMC)

- Particles evolve prior $\rightarrow$ posterior through a sequence of intermediate densities p_t : reweight, resample, MCMC move
- Highly parallelizable: GPU acceleration
- What choice for sequence p_t ?



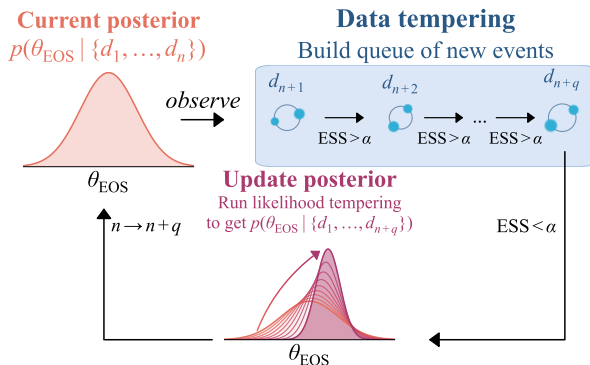
Likelihood vs. data tempering

- **Likelihood tempering**: anneal in the full likelihood, $\mathcal{L}(\theta)^{\beta_t}$, $\beta_t : 0 \rightarrow 1$
 - Used in GW PE (POCOMC [2, 3], ASPIRE [4]) and JESTER [5]
- **Data tempering**: anneal in new observations one at a time
 - Reuse previous posterior



Adaptive hybrid algorithm

- New events **queued** while effective sample size (ESS) stays high
- ESS drops below threshold $\alpha \Rightarrow$ **update** posterior with likelihood tempering over the batch
- Naturally handles **growing catalogs**, without restarting from the prior

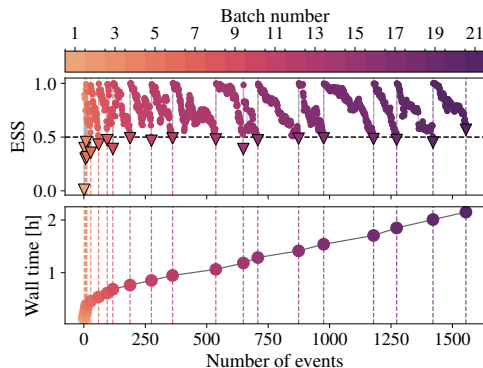


Simulating one month of ET data

Demonstrated with proof of concept:

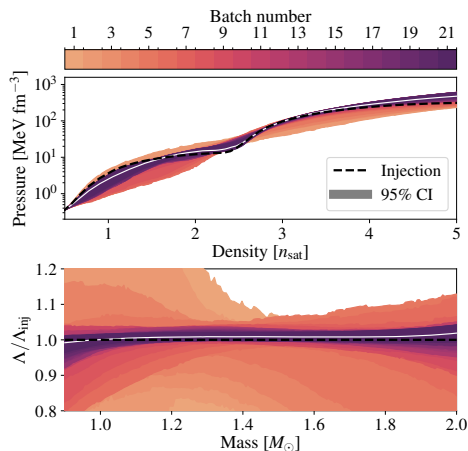
- ET 2L-misaligned configuration, Gaussian noise, merger rate $106.6 \text{ Gpc}^{-3}\text{yr}^{-1}$
- $\text{SNR} > 12 \Rightarrow 1555$ detected BNS sources in 1 month
- Single-event PE: nested sampling, masses + tidal deformabilities only, $f_{\min} = 20 \text{ Hz}$, TaylorF2 (simplified settings!)
- JESTER:
 - Sample metamodel + CSE parametrization
 - $N = 2000$ SMC particles, ESS threshold $\alpha = 50\%$, prior informed by PSR J0740+6620 [6]

SMC diagnostics



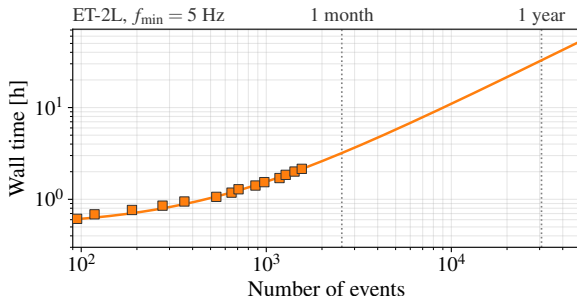
- ESS **below 50%** triggers an update (∇)
- Early events highly informative: frequent updates
- Later batches: hundreds of events before the next update
- Total: 22 updates, $\sim$ **2.2h** on 1 NVIDIA H100 GPU

Equation of state constraints



- Bulk region ($1-3 n_{\text{sat}}$): pressure under- or overestimated
- Injected Λ within the 95% credible band throughout
- Final batch: $\sim 1\%$ constraints on Λ across probed masses

Extrapolating to a full year



- Compute becomes memory-bound $\Rightarrow$ linear fit ($N > 400$)
- More realistic settings ($f_{\min} = 5$ Hz): ~ 2559 events/month, $\sim 30\,000$ /year
- Projected runtime: **a few hours** (1 month), **~ 1.5 days** (1 year) on a single NVIDIA H100 GPU

Conclusion & outlook

- Recipe for **scalable**, **sequential** hierarchical inference:
 - **GPU**: fast hyperlikelihoods
 - **SMC**: data tempering sequentially updates posteriors
- Demonstrated on a $\mathcal{O}(10^3)$ -event ET mock catalog (nested sampling)
- Extrapolation: a **full year** ($\sim 30\,000$ events) feasible within days on a single GPU
- Outlook: joint EOS + population + cosmology inference [5]

Thanks for listening!



References I

- [1] Michele Maggiore et al. “Science Case for the Einstein Telescope”. In: *JCAP* 03 (2020), p. 050. DOI: [10.1088/1475-7516/2020/03/050](https://doi.org/10.1088/1475-7516/2020/03/050). arXiv: [1912.02622](https://arxiv.org/abs/1912.02622) [astro-ph.CO].
- [2] Minas Karamanis et al. “pocoMC: A Python package for accelerated Bayesian inference in astronomy and cosmology”. In: *J. Open Source Softw.* 7.79 (2022), p. 4634. DOI: [10.21105/joss.04634](https://doi.org/10.21105/joss.04634). arXiv: [2207.05660](https://arxiv.org/abs/2207.05660) [astro-ph.IM].
- [3] Michael J. Williams et al. “Validating sequential Monte Carlo for gravitational-wave inference”. In: *Mon. Not. Roy. Astron. Soc.* 543.2 (2025), pp. 1479–1493. DOI: [10.1093/mnras/staf1458](https://doi.org/10.1093/mnras/staf1458). arXiv: [2506.18977](https://arxiv.org/abs/2506.18977) [astro-ph.IM].
- [4] Michael J. Williams. “Accelerated Sequential Posterior Inference via Reuse for Gravitational-Wave Analyses”. In: *Phys. Rev. Lett.* 137.2 (2026), p. 021410. DOI: [10.1103/5hvn-3lmh](https://doi.org/10.1103/5hvn-3lmh). arXiv: [2511.04218](https://arxiv.org/abs/2511.04218) [hep-ex].
- [5] Hauke Koehn et al. “Binary neutron stars in the next-generation era: Multi-messenger detection prospects and constraints on the equation of state, mass distribution, and cosmology”. In: (July 2026). arXiv: [2607.28438](https://arxiv.org/abs/2607.28438) [astro-ph.HE].
- [6] E. Fonseca et al. “Refined Mass and Geometric Measurements of the High-mass PSR J0740+6620”. In: *Astrophys. J. Lett.* 915.1 (2021), p. L12. DOI: [10.3847/2041-8213/ac03b8](https://doi.org/10.3847/2041-8213/ac03b8). arXiv: [2104.00880](https://arxiv.org/abs/2104.00880) [astro-ph.HE].

References II

- [7] Peter Neal and Gareth Roberts. “Optimal scaling of random walk Metropolis algorithms with non-Gaussian proposals”. In: *Methodology and Computing in Applied Probability* 13.3 (2011), pp. 583–601.

Backup: SMC — likelihood tempering

$$p_t(\theta) = \frac{\mathcal{L}(\theta)^{\beta_t} \pi(\theta)}{\mathcal{Z}_t}, \quad \beta_t = \sup_{\beta \in [\beta_{t-1}, 1]} \{ \text{ESS}_t(\beta) \geq \alpha N \}$$

```
1:  $t \leftarrow 1, \beta_1 \leftarrow 0, \hat{\mathcal{Z}}_1 \leftarrow 1$ 
2: for  $i = 1, \dots, N$  do
3:    $\theta_1^i \sim \pi(\theta)$  ▷ sample from the prior
4: end for
5: while  $\beta_t < 1$  do
6:    $t \leftarrow t + 1$ 
7:    $\beta_t \leftarrow \sup_{\beta \in [\beta_{t-1}, 1]} \{ \text{ESS}_t(\beta) \geq \alpha N \}$  ▷ adaptive temperature
8:   for  $i = 1, \dots, N$  do
9:      $w_t^i \leftarrow \mathcal{L}(\theta_{t-1}^i)^{\beta_t - \beta_{t-1}}$  ▷ reweight
10:  end for
11:   $\hat{\mathcal{Z}}_t \leftarrow \hat{\mathcal{Z}}_{t-1} \times \frac{1}{N} \sum_i w_t^i$  ▷ update evidence
12:  resample  $\{\theta_{t-1}^i\}$  with weights  $\{w_t^i\}$ 
13:  move particles with MCMC kernel  $\mathcal{K}_t$  ▷ rejuvenation
14: end while
```

Settings: $N = 2000$, $\alpha = 0.9$, Gaussian random-walk kernel, 20 MCMC steps, target acceptance 0.234 [7].

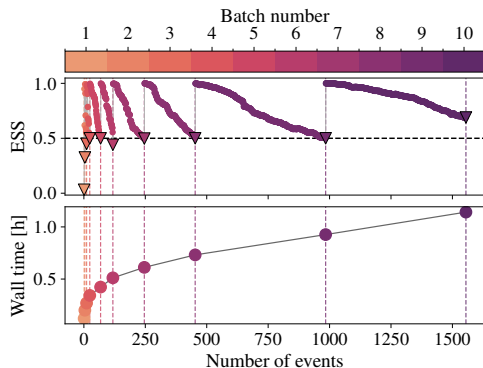
Backup: SMC — data tempering

$$\mathcal{P}_t(\theta) = \frac{\mathcal{L}(d_{1:b_t}|\theta)\pi(\theta)}{\mathcal{Z}_t}, \quad b_t = \sup_{b \in [b_{t-1}, T]} \{\text{ESS}_t(b) \geq \tilde{\alpha}N\}$$

```
1:  $t \leftarrow 0, n \leftarrow 0, q \leftarrow 0, \hat{\mathcal{Z}}_0 \leftarrow 1$ 
2: for  $i = 1, \dots, N$  do
3:    $\theta_0^i \sim \pi(\theta), w_0^i \leftarrow 1/N$ 
4: end for
5: while  $t < M$  do
6:    $t \leftarrow t + 1, q \leftarrow q + 1$ 
7:   for  $i = 1, \dots, N$  do
8:      $w_t^i \leftarrow w_{t-1}^i \times \mathcal{L}_t(\theta_n^i)$  ▷ reweight with new event
9:   end for
10:  if  $\text{ESS}_t < \tilde{\alpha}N$  or  $t = M$  then
11:     $\{\theta_t^i\}, \hat{\mathcal{Z}}_{n:n+q} \leftarrow \text{Alg. 1}(\mathcal{P}_n, \mathcal{L}_{n:n+q})$  ▷ batch update
12:     $\hat{\mathcal{Z}}_t \leftarrow \hat{\mathcal{Z}}_n \times \hat{\mathcal{Z}}_{n:n+q}, n \leftarrow n + q, q \leftarrow 0$ 
13:     $w_t^i \leftarrow 1/N$  ▷ clear the queue
14:  end if
15: end while
```

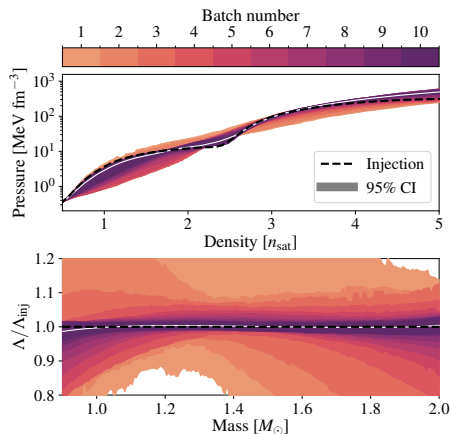
Threshold $\tilde{\alpha} = 0.5$; inner likelihood-tempering step uses the same settings as Algorithm 1.

Backup: zero-noise SMC diagnostics



- Same catalog, **no detector noise**
- ESS decays smoothly and monotonically, no erratic drops
- Fewer update steps needed: **10** batches vs. 22 with noise
- Lower total wall time as a result

Backup: zero-noise EOS constraints



- Same catalog, **no detector noise**
- Median Λ curve matches the injection much more closely
- No slight bias in $\Lambda(M)$